

Parametrization of Fractions in an Interval

Topic.

Number theory, coprime fractions, parametrization.

Statement.

Let M be the set of irreducible fractions with positive integer numerator and denominator lying between

$$\frac{3}{5} \quad \text{and} \quad \frac{2}{3}.$$

Prove that

$$u, v \in \mathbb{N}, \quad \gcd(u, v) = 1 \implies \frac{3u + 2v}{5u + 3v} \in M.$$

Also prove that every fraction from M can be represented in the form

$$\frac{3u + 2v}{5u + 3v}$$

for some

$$u, v \in \mathbb{N}, \quad \gcd(u, v) = 1.$$

Solution.

1.

Let

$$u, v \in \mathbb{N}, \quad \gcd(u, v) = 1.$$

Put

$$\frac{p}{q} = \frac{3u + 2v}{5u + 3v},$$

where

$$p = 3u + 2v, \quad q = 5u + 3v.$$

First prove that

$$\frac{3}{5} < \frac{p}{q} < \frac{2}{3}.$$

We have

$$\frac{p}{q} > \frac{3}{5} \iff 5p > 3q.$$

Substitute p and q :

$$5(3u + 2v) > 3(5u + 3v)$$

$$15u + 10v > 15u + 9v$$

$$v > 0.$$

This is true since

$$v \in \mathbb{N}.$$

Also,

$$\frac{p}{q} < \frac{2}{3} \iff 3p < 2q.$$

Substitute p and q :

$$3(3u + 2v) < 2(5u + 3v)$$

$$9u + 6v < 10u + 6v$$

$$u > 0.$$

This is true since

$$u \in \mathbb{N}.$$

Therefore,

$$\frac{3}{5} < \frac{3u + 2v}{5u + 3v} < \frac{2}{3}.$$

Now prove that the fraction is irreducible.

Let

$$d \mid (3u + 2v), \quad d \mid (5u + 3v).$$

Then d divides any integer linear combination of these two numbers.

Hence

$$d \mid (3(5u + 3v) - 5(3u + 2v)).$$

But

$$3(5u + 3v) - 5(3u + 2v) = 15u + 9v - 15u - 10v = -v.$$

Therefore,

$$d \mid v.$$

Also,

$$d \mid (2(5u + 3v) - 3(3u + 2v)).$$

But

$$2(5u + 3v) - 3(3u + 2v) = 10u + 6v - 9u - 6v = u.$$

Therefore,

$$d \mid u.$$

Thus,

$$d \mid u, \quad d \mid v.$$

Since

$$\gcd(u, v) = 1,$$

we get

$$d = 1.$$

Hence

$$\gcd(3u + 2v, 5u + 3v) = 1.$$

So the fraction is irreducible and lies in

$$\left(\frac{3}{5}, \frac{2}{3}\right).$$

Therefore,

$$\frac{3u + 2v}{5u + 3v} \in M.$$

2.

Let

$$\frac{p}{q} \in M.$$

Then

$$\frac{3}{5} < \frac{p}{q} < \frac{2}{3}, \quad \gcd(p, q) = 1.$$

Define

$$u = 2q - 3p, \quad v = 5p - 3q.$$

Prove that

$$u, v \in \mathbb{N}.$$

From

$$\frac{p}{q} < \frac{2}{3}$$

we get

$$3p < 2q.$$

Hence

$$2q - 3p > 0,$$

so

$$u > 0.$$

From

$$\frac{p}{q} > \frac{3}{5}$$

we get

$$5p > 3q.$$

Hence

$$5p - 3q > 0,$$

so

$$v > 0.$$

Therefore,

$$u, v \in \mathbb{N}.$$

Now compute the numerator:

$$3u + 2v = 3(2q - 3p) + 2(5p - 3q).$$

Then

$$3u + 2v = 6q - 9p + 10p - 6q = p.$$

Compute the denominator:

$$5u + 3v = 5(2q - 3p) + 3(5p - 3q).$$

Then

$$5u + 3v = 10q - 15p + 15p - 9q = q.$$

Therefore,

$$\frac{3u + 2v}{5u + 3v} = \frac{p}{q}.$$

It remains to prove that

$$\gcd(u, v) = 1.$$

Let

$$d \mid u, \quad d \mid v.$$

Then

$$d \mid (3u + 2v), \quad d \mid (5u + 3v).$$

But

$$3u + 2v = p, \quad 5u + 3v = q.$$

Hence

$$d \mid p, \quad d \mid q.$$

Since

$$\gcd(p, q) = 1,$$

we get

$$d = 1.$$

Therefore,

$$\gcd(u, v) = 1.$$

Thus,

$$\forall \frac{p}{q} \in M \quad \exists u, v \in \mathbb{N} : \quad \gcd(u, v) = 1 \quad \text{and} \quad \frac{p}{q} = \frac{3u + 2v}{5u + 3v}.$$

This proves the required statement.